Hydrogen poles

Walter Russell's "four pairs of poles — not one pair" read as hydrogen: one atom's magnetic poles level by level, the four magnets of H_2, and hydrogen's lines as kronspice resonances (register entry H34)

kronspice simulation report · 2026-11-07 · recorded run: run_output.txt (about a minute)

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Summary

Hypothesis H34 (user hypothesis, 2026-11-05) reads Walter Russell's painting "Analysis of the Octave Bar Magnet", beneath which he wrote *"a bar magnet has four pairs of poles — not one pair"*, as hydrogen, and takes the painting's two touching spheres as two hydrogen atoms (the bar reading is H33, *octave_magnet*). The example computes one atom's magnetic field level by level from the electron's orbital current and spin magnetization, counts its N and S zones as a bench would, builds H_2 with the Heitler–London wavefunction and its four spin magnets, and finds hydrogen's spectral lines as resonances of circuits solved by kronspice. This PDF covers the example's three reports: one atom (Parts 1–2), two atoms (Part 3) and the spectrum (Part 4).

- **Ground state: one pair.** 1s½ has a dipole of −1.001160 μ_B (equal to the Landé value) and no octupole; so does every j = ½ level.
- **Four pairs first appear in 2p3/2.** The m = ½ state has octupole −54.062621 $\mu_B \cdot a0^2$ and shows four N S N S zones round the atom from 0.5 to 15.4 a0 (r_c = 15.61 a0); the octupole comes entirely from the electron's spin magnetization.
- **Selection rules hold numerically:** Wigner–Eckart ratio −3.000000; even-l moments and j = ½ octupoles vanish to round-off; series field = direct sum of 1.28 million current loops to 2.4×10^{-4}.
- **H_2: the bond cancels the electron magnets.** The singlet binds (R_e 1.6425 a0, D_e 3.1557 eV; measured 1.4011 a0, 4.748 eV); the triplet repels at every R. The bound molecule has zero spin density, leaving only the proton dipoles (1.52×10^{-3} μ_B each).
- **Spectrum as resonances:** Lyman, Balmer, Paschen and Brackett α within 16 ppm, the 2p fine structure 10968.90 MHz against 10969.10 MHz, 21 cm +56 ppm and ESR +18 ppm from the model; H_2/HD/D_2 vibration 13.5 % low with isotope ratios to 0.07 %.
- **Status: bounded.** No measurement of an excited hydrogen atom's magnetic octupole, or of H_2's nuclear four-zone field, is known.

What is being tested

Hypothesis (H34). Russell's "four pairs of poles — not one pair" describes hydrogen, and his two touching spheres are two hydrogen atoms. **Standard prediction.** Dirac/QED hydrogen with these multipoles: ground-state hydrogen has one pair of poles (a dipole); four pairs (an octupole) first appear in the 2p3/2 level. In H_2 the bond pairs the electron spins, so only the protons' tiny dipoles remain.

What "four pairs" means for an atom

- **Poles come in powers of two.** A dipole has one pair; an octupole (2^3 poles) has four. "Four pairs, not one pair" names the octupole against the dipole.
- **No two-pair state.** A state of definite parity has no magnetic quadrupole, so an atom's magnetic poles come as one pair or four pairs, never two.
- **Four pairs need higher states.** A magnetic 2^k-pole needs $2j \geq k$. The ground state (1s, j = ½) can have only one pair; four pairs first become possible in 2p3/2.
- **On a sphere round the atom** an axial octupole shows four alternating zones of B_r (N S N S from pole to pole); a dipole shows two.

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Report 1 — One hydrogen atom (Parts 1–2)

The model

- **The electron:** Pauli spinors |n l j m> built with Clebsch–Gordan coefficients from the hydrogen radial functions. Its field has two sources: the orbital current $J = -2\mu_B \Sigma\, m_l|\psi|^2/\rho$, and the spin magnetization $M = -(g_s/2)\mu_B\, \psi^\dagger\sigma\psi$ with $g_s = 2.002319$.
- **The field:** both enter as one azimuthal current, J_{orb} + curl M, whose field is computed exactly from the Legendre-series Green's function (Jackson §5.5).
- **The proton:** its spin dipole ($+1.5210\times10^{-3}\ \mu_B$, $m_l = +½$) is added at the centre.

The atom's magnetism, level by level

Level	$\int\|\psi\|^2$	Dipole (μ_B)	Landé $-g_j m_j$	Octupole ($\mu_B \cdot a0^2$)	Pole pattern round the atom
1s½, m = ½	1.000000	−1.001160	−1.001160	0	2 zones (one pair) at every radius
2p½, m = ½	1.000000	−0.332947	−0.332947	0	2 zones at every radius
2p3/2, m = 3/2	1.000000	−2.001160	−2.001160	+18.020874	2 zones at every radius (r_c = 4.25 a0 falls inside the cloud)
2p3/2, m = ½	1.000000	−0.667053	−0.667053	−54.062621	**4 zones 0.5–15.4 a0**, then 2 (r_c = 15.61 a0)

Split by source (dipole, octupole):

Level	Orbital current	Spin magnetization
1s½, m = +½	−0.000000, −0.000000	−1.001160, +0.000000
2p½, m = +½	−0.666667, −0.000000	+0.333720, +0.000000
2p3/2, m = +3/2	−1.000000, −0.000000	−1.001160, +18.020874
2p3/2, m = +½	−0.333333, −0.000000	−0.333720, −54.062621

The octupole comes entirely from the electron's spin magnetization; the l = 1 orbital current carries none. The j = ½ levels have none in either part, as 2j ≥ 3 requires.

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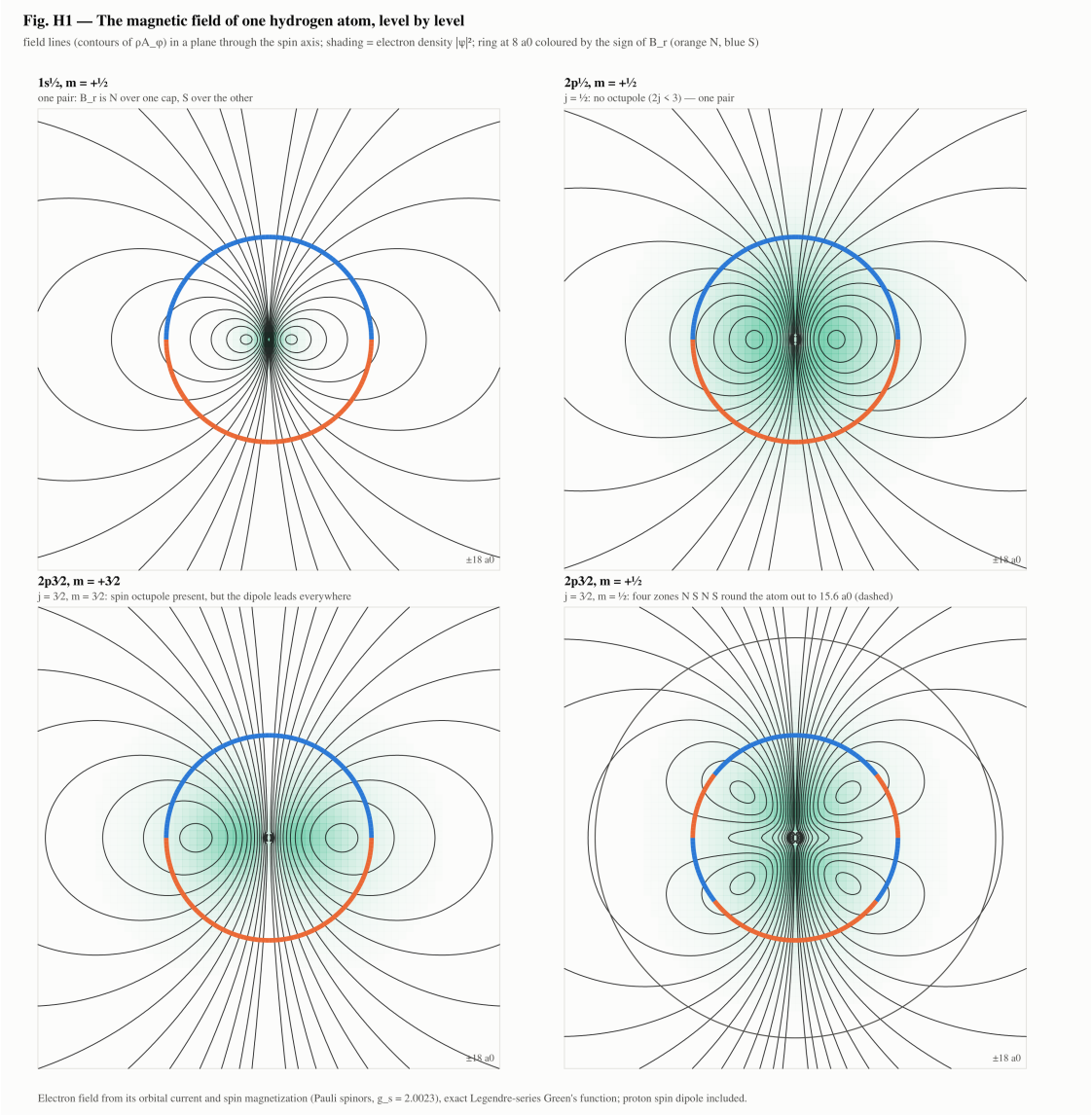


Figure 1 (H1). Field lines (contours of ρA_φ) of each level in a plane through the spin axis, over the electron density; the ring at 8 a0 is coloured by the sign of B_r (orange N, blue S). Panels: 1s½ m = +½, 2p½ m = +½, 2p3/2 m = +3/2, 2p3/2 m = +½. Only the last shows four zones; its dashed circle marks 15.6 a0, where they end.

Counting the poles

The pole count is read on spheres of growing radius; each sign change of B_r from pole to pole separates an N zone from an S zone. Outside the cloud the octupole leads inside $r_c = \sqrt{(3m_3/m_1)}$, or $\sqrt{(-2m_3/m_1)}$ when the ratio is negative; beyond it the dipole leads, because it falls as $1/r^3$ and the octupole as $1/r^5$. For m = 3/2, r_c = 4.25 a0 lies inside the cloud, where the field is not the exterior expansion, and no four-zone pattern appears at any radius.

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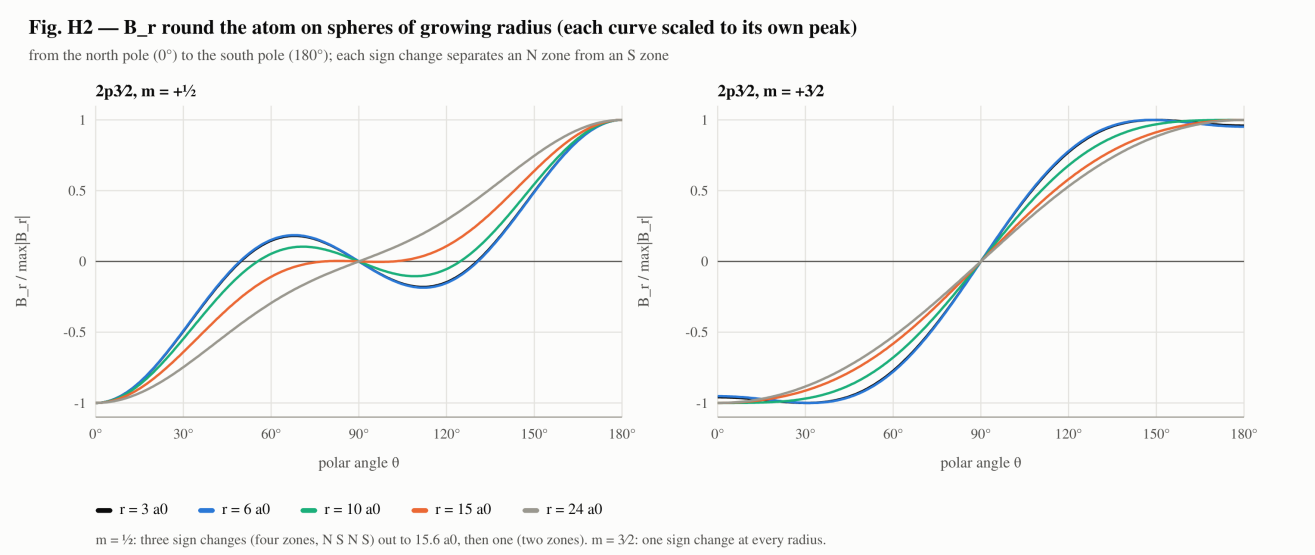


Figure 2 (H2). B_r from the north pole (0°) to the south pole (180°) on spheres of 3 to 24 a0, each curve scaled to its own peak. Left, 2p3/2 m = +½: three sign changes (four zones) out to 15.6 a0, then one. Right, m = +3/2: one sign change at every radius.

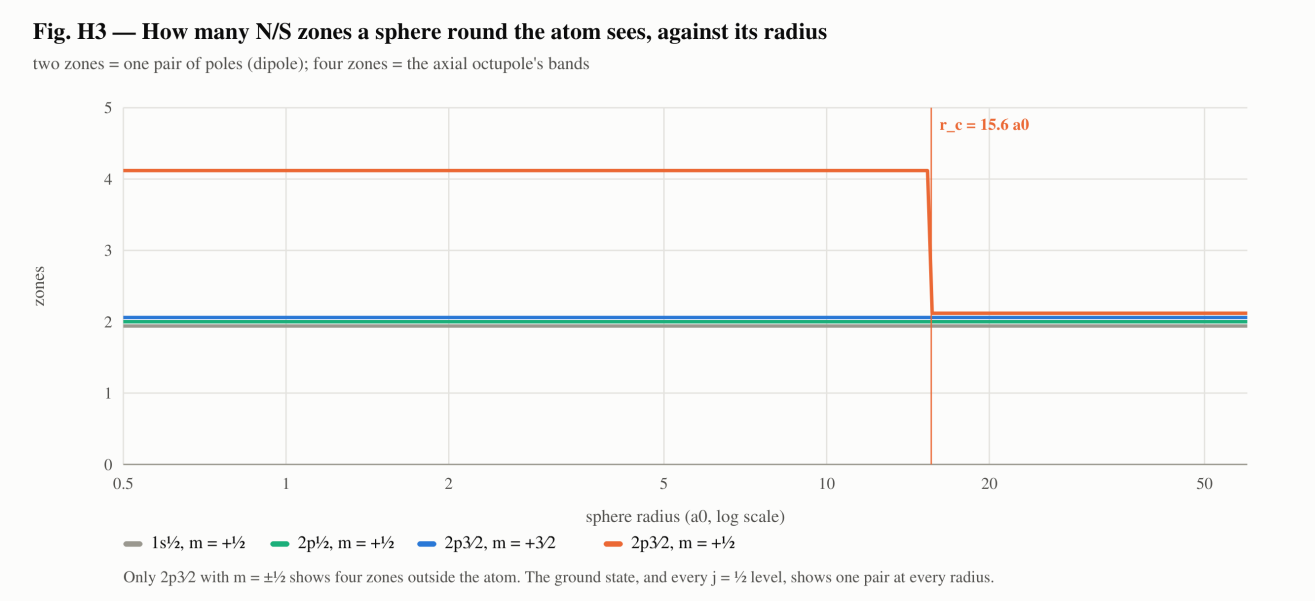


Figure 3 (H3). Zones against sphere radius (log scale). Two zones (one pair) for the ground state and every j = ½ level; four for 2p3/2 m = ½ out to r_c = 15.6 a0.

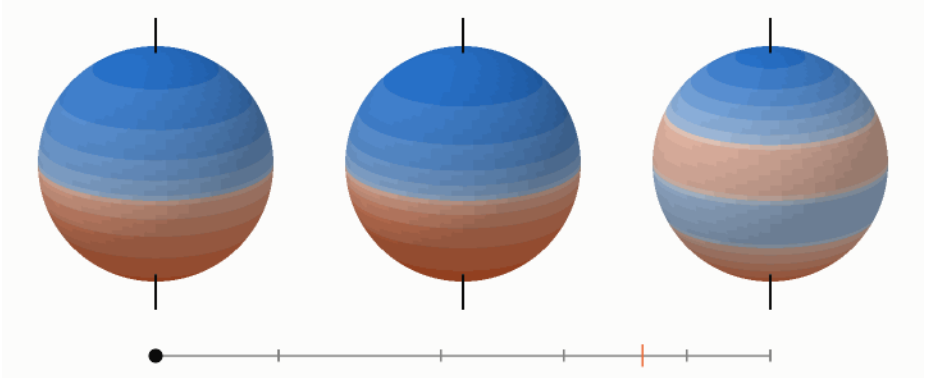


Figure 4. First frame (radius 1 a0) of the animation viz/atom_spheres.gif: spheres round 2p½ (left), 2p3/2 m = 3/2 (middle) and 2p3/2 m = ½ (right), coloured by B_r (orange N, blue S). The right-hand sphere carries four bands; in the animation they stay until the radius passes 15.6 a0 (the orange tick), then become two.

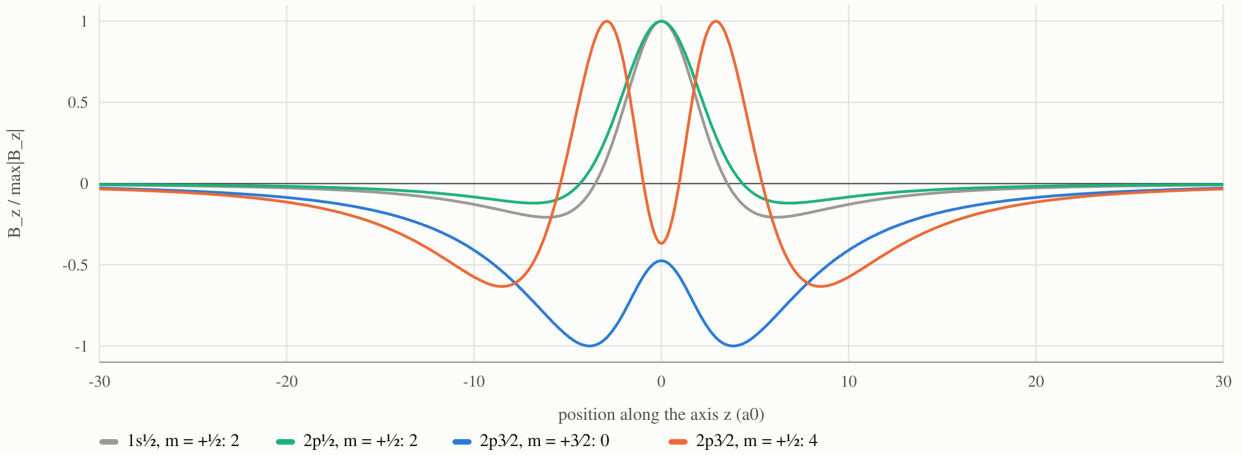
The octave bar's own bench scan (*octave_magnet*) on the atom: sign changes of B_z on a line beside the spin axis, z from −60 to 60 a0. A dipole gives 2; more mean more pole sites along the line.

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Level	ρ = 2 a0	ρ = 5 a0	ρ = 10 a0	ρ = 20 a0
1s½, m = +½	2	2	2	2
2p½, m = +½	0	2	2	2
2p3/2, m = +3/2	0	0	2	2
2p3/2, m = +½	0	**4**	2	2

Fig. H4 — The octave bar's bench test on the atom: B_z along a line 5 a0 beside the spin axis

the scan of examples/octave_magnet: a dipole gives two sign changes; more mean more pole sites along the line



Sign changes in the legend. 2p3⁄2 m = ½ gives four at this distance: two N and two S sites along the line, the four-zone pattern seen from the side.

Figure 5 (H4). B_z along a line 5 a0 beside the spin axis, each curve scaled to its own peak. Sign changes: 1s½ 2, 2p½ 2, 2p3/2 m = +3/2 0, 2p3/2 m = +½ 4 (two N and two S sites along the line).

Checks

Check	Result
Normalization $\int\|\psi\|^2$, every level	1 to 10^{-8}
Dipole against the Landé value $-g_j m_j \mu_B$	equal to 10^{-8}
Even-l moments (parity)	≤ 4.2×10^{-10}
Octupole of the j = ½ levels (2j ≥ k)	≤ 10^{-11}
Wigner–Eckart: octupole(m = ½)/octupole(m = 3/2), theory −3 (proportional to m(5m² − 3j(j+1) + 1))	−3.000000
Series field against a direct sum of 1 280 000 current loops, 4 points outside the cloud	worst 2.4×10^{-4} of |B|

Report 2 — Two hydrogen atoms, four magnets (Part 3)

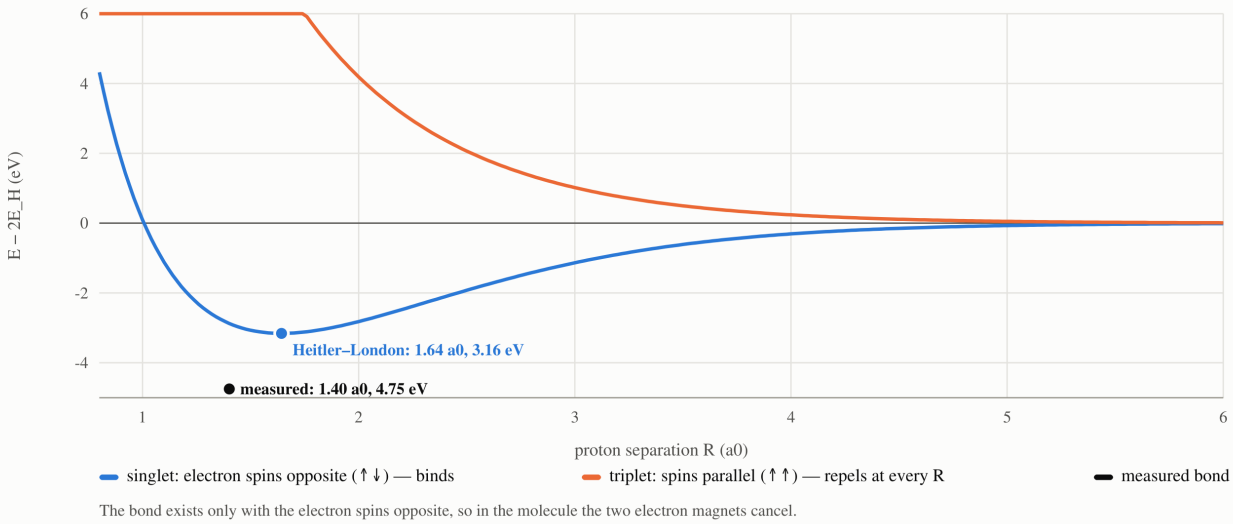
Russell's two touching spheres read as two hydrogen atoms bonding into H_2. The molecule holds four small magnets: two electron spins and two proton spins. The model is the Heitler–London wavefunction (Heitler & London 1927) with the exact one- and two-centre integrals (Sugiura 1927 for the exchange integral).

- **Singlet (spins opposite) binds:** R_e 1.6425 a0 (0.8692 Å), D_e 3.1557 eV, the classic Heitler–London result; measured 1.4011 a0 and 4.748 eV. The simple wavefunction gets two thirds of the binding.
- **Triplet (spins parallel) repels at every R:** $E - 2E_H$ = +10.099 eV at 1.4 a0, +4.189 eV at 2 a0, +0.2392 eV at 4 a0.

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Fig. M1 — Why the two electron magnets pair off: the Heitler–London energies of H_2

energy of two hydrogen atoms against their separation, relative to the separated atoms

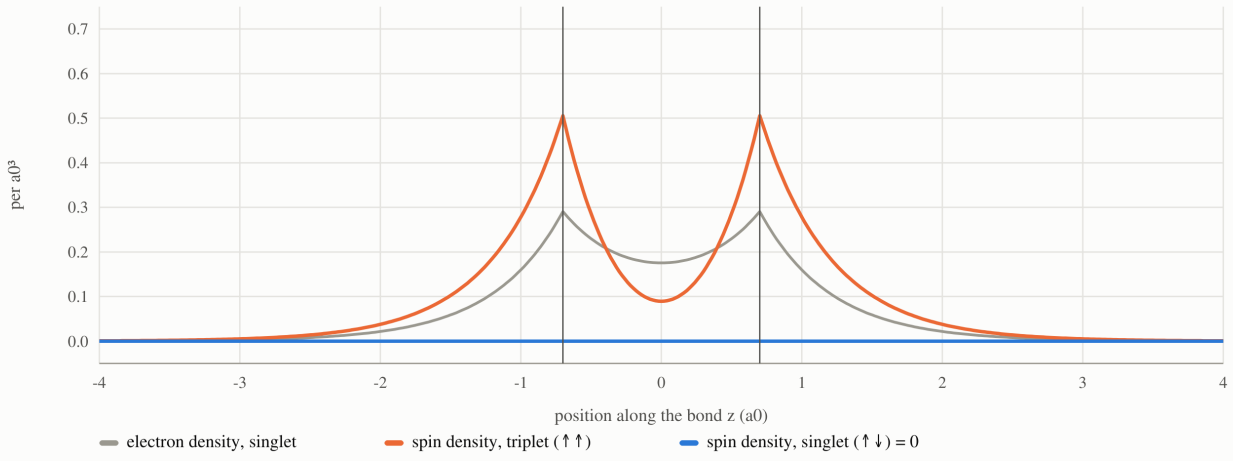


The bond exists only with the electron spins opposite, so in the molecule the two electron magnets cancel.

Figure 6 (M1). Heitler–London energies of two hydrogen atoms against their separation, relative to the separated atoms. Only the singlet (blue) has a well, at 1.64 a0 and 3.16 eV; the measured bond (1.40 a0, 4.75 eV) is marked. The triplet (orange) repels at every R.

Fig. M2 — Electron density and spin density along the bond axis (R = 1.40 a0)

the spin density is what makes the electrons magnetic; in the bound (singlet) molecule it is zero everywhere



Dashed lines: the protons. The triplet's spin density sits on both atoms (cusps at the protons); the singlet's vanishes identically.

Figure 7 (M2). Along the bond axis at R = 1.40 a0: electron density of the singlet (grey), spin density of the triplet (orange, cusps at the protons) and of the singlet (blue, zero everywhere, because its spin part gives $\langle\sigma_z\rangle = 0$ for each electron).

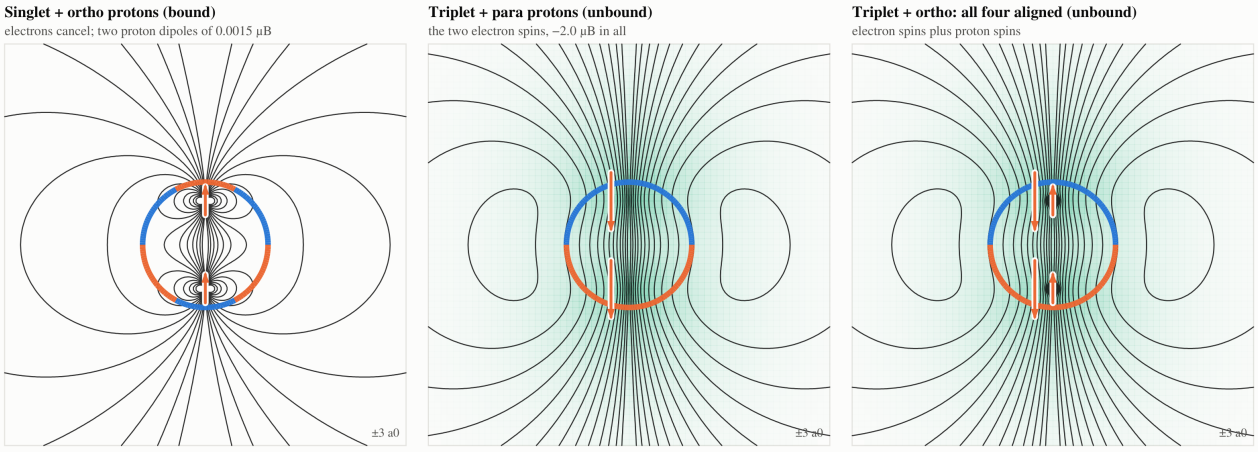
The field of the four magnets, at the measured bond length 1.4011 a0 with spins quantized along the bond. The electron field comes from the triplet spin magnetization $M_z = -(g_s/2)\mu_B\, s_z$, summed as current loops and evaluated inside the cloud on cell corners, where the neighbouring loops' near fields cancel (grid h = 0.05 against 0.025: worst 3.4×10^{-4} of |B|; h = 0.1 against 0.05: 1.4×10^{-3}). The proton spins are point dipoles at the nuclei.

Spin arrangement	∫ spin density	Moment (μ_B)	Bound?	Zones round the molecule	B_z sign changes at ρ = 2 a0
singlet electrons + para protons	0.0000	+0.00000	yes	none — no magnetic field at all	—
singlet electrons + ortho protons	0.0000	+0.00304	yes	2 zones 0.20–0.29 a0; 4 zones 0.31–1.56 a0; 2 zones 1.62–20.00 a0	2
triplet electrons + para protons	2.0001	−2.00232	no	2 zones 0.20–20.00 a0	2
triplet electrons + ortho protons (all four aligned)	2.0001	−1.99928	no	2 zones 0.20–0.59 a0; 4 zones 0.61–0.77 a0; 2 zones 0.81–20.00 a0	2

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Fig. M3 — The field of H_2's four magnets, in a plane through the bond (bond vertical)

field lines (contours of ρA_φ); shading = spin density; ring at 1 a0 coloured by the sign of B_r (orange N, blue S); arrows = the magnetic moments (orange)



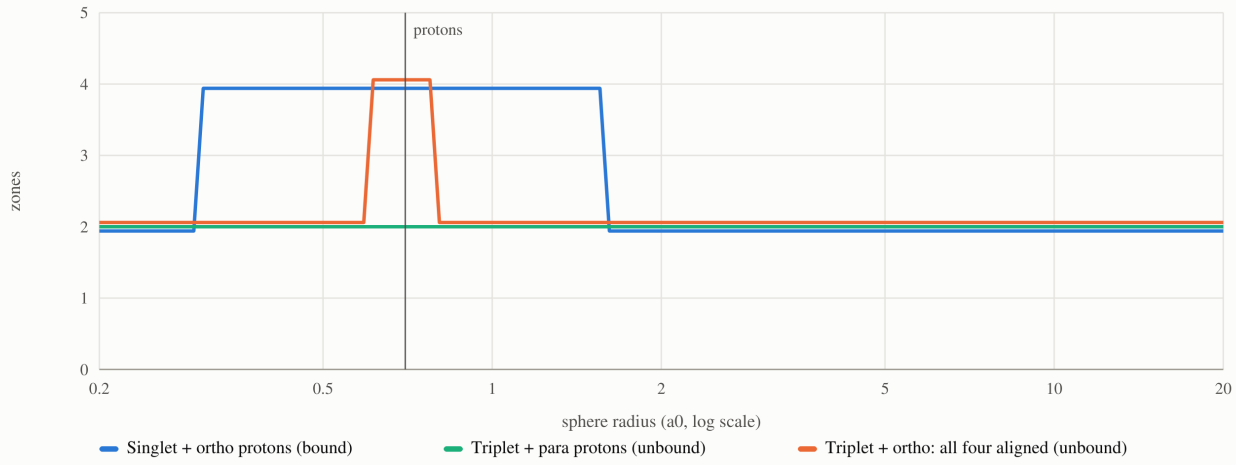
Long arrows, beside the axis: electron spin moments (opposite to the spins, the electron being negative). Short arrows, on the axis: proton moments (drawn far larger than their 1/658 share). In the bound singlet only the protons remain, and the para form (proton spins opposite) has no field at all.

Zones round the molecule (Fig. M4): ortho protons alone show four zones between 0.3 and 1.6 a0 — two dipoles end to end, each 658 times weaker than one electron's. With the triplet electrons the field is one pair at every radius except a thin shell round the protons (0.6–0.8 a0).

Figure 8 (M3). Field lines (contours of ρA_φ) in a plane through the bond (bond vertical), over the spin density; ring at 1 a0 coloured by the sign of B_r. Panels: singlet + ortho protons (bound; electrons cancel, two proton dipoles of 0.0015 μ_B), triplet + para protons (unbound; −2.0 μ_B in all), triplet + ortho (all four aligned, unbound). Long arrows are the electron moments, short arrows the protons' (drawn far larger than their 1/658 share).

Fig. M4 — N/S zones on spheres round the H_2 molecule, against their radius

centred between the protons (each at 0.70 a0 from the centre)



Singlet + para (bound, the commonest form at low temperature) has no field and no zones; it is left off the chart.

Figure 9 (M4). Zones on spheres centred between the protons (each 0.70 a0 from the centre), against radius. Ortho protons alone make four zones near the molecule (two dipoles end to end); the triplet electrons make one pair at almost every radius. Singlet + para has no field and no zones and is left off.

The two-atom picture does hold four small magnets, and with all four aligned the molecule would carry about 2 μ_B. But that needs the electron spins parallel, and the triplet does not bind. The bond forms only with the spins opposite, and then the electron magnets cancel exactly, at every point. What is left are the proton magnets, 658 times weaker. Ortho-H_2 shows a four-zone pattern near the protons; para-H_2, the commonest form at low temperature, has no magnetic field at all. The rotating molecule's own small moment (ortho, J = 1) is not modelled.

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Report 3 — Hydrogen's spectral lines as circuit resonances (Part 4)

- **Atomic levels** are resonances of Kron's 1945 radial network (*Phys. Rev.* 67, 39), one network per orbital (1s, 2p, 3d, 4f, 5g), solved by kronspice's AC analysis in its dual form as in *light_hydrogen*: each level is where the driving-point reactance changes sign. Grids of h = 0.1 and 0.05 a0 are Richardson-extrapolated, and energies are scaled by the reduced mass μ/m = 0.999456.
- **Fine structure:** the p networks carry the spin–orbit term $(g_s - 1)(\alpha^2/2)\langle L\cdot S\rangle/r^3$ (Bethe & Salpeter §12), so the j = ½ and j = 3/2 networks resonate apart.
- **H_2 vibration:** an LC circuit, the masses as capacitors and the Heitler–London spring (k = 0.2759 $E_h/a0^2$ at 1.6425 a0) as an inductor 1/k; its single transfer pole is the vibration.
- **Two lines come from the model, not a circuit:** the 21 cm line uses the computed ψ(0) of Part 1 (Fermi contact), and the ESR frequency the computed dipole.

Level	kronspice h	kronspice h/2	Richardson	Exact	AC solves
1s	−0.4987562112	−0.4996878900	−0.4999984496	−0.500000	112
2p (no spin–orbit)	−0.1250260815	−0.1250065129	−0.1249999900	−0.125000	108
2p½	−0.1250271946	−0.1250076252	−0.1250011021	−0.125000	108
2p3/2	−0.1250255250	−0.1250059568	−0.1249994341	−0.125000	108
3d	−0.0555565843	−0.0555558128	−0.0555555556	−0.055556	104
4f	−0.0312501395	−0.0312500349	−0.0312500000	−0.031250	104
5g	−0.0200000310	−0.0200000072	−0.0199999993	−0.020000	102

Energies in hartree. Twin: kronspice resonance against the network matrix (Sturm), h/2: worst 4.7×10^{-14} Ha. The fine structure 2p3/2 − 2p½ from the two networks' resonances is 10979.6 MHz at h, 10971.6 MHz at h/2 and 10968.9 MHz by Richardson; with the spin–orbit term off the two are one network and one resonance.

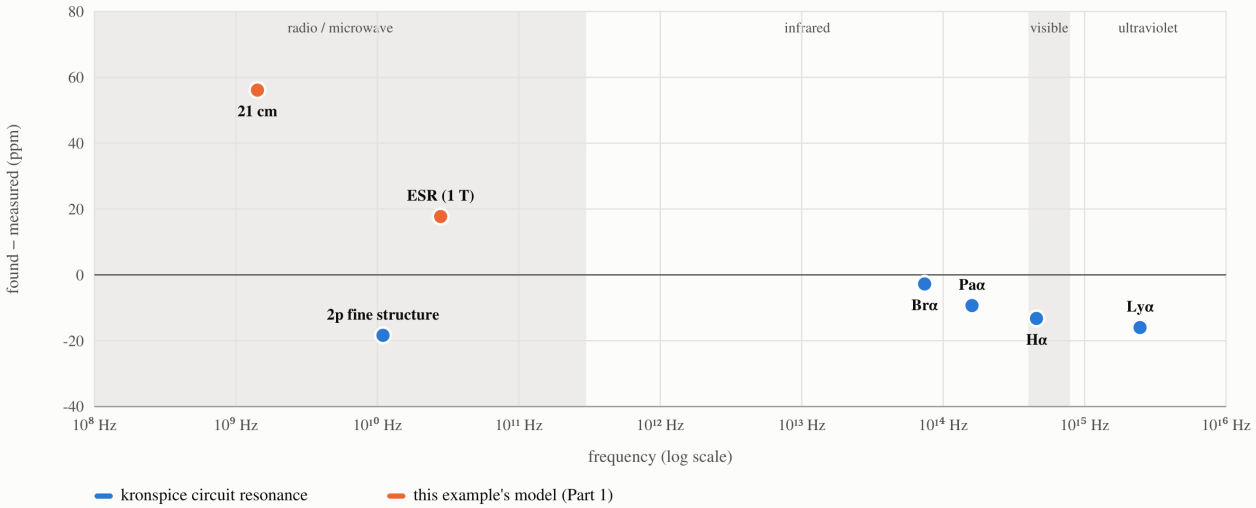
Line	Found	Measured	Difference	How
Lyman α (2p → 1s)	121.5689 nm	121.5670 nm	−0.0016 %	kronspice resonance
Balmer α, Hα (3d → 2p)	656.4697 nm	656.4610 nm	−0.0013 %	kronspice resonance
Paschen α (4f → 3d)	1875.6274 nm	1875.6100 nm	−0.0009 %	kronspice resonance
Brackett α (5g → 4f)	4052.2813 nm	4052.2700 nm	−0.0003 %	kronspice resonance
fine structure 2p3/2 − 2p½	10968.90 MHz	10969.10 MHz	−0.0018 %	kronspice resonance
hyperfine 1s, 21 cm (F = 1 → 0)	1420.4855 MHz	1420.4058 MHz	+0.0056 %	model (Fermi contact, ψ(0) of Part 1)
electron spin resonance of H, per tesla	28024.95 MHz	28024.45 MHz	+0.0018 %	model (dipole of Part 1)

The remainders are physics these models leave out: for the optical lines the Dirac level shifts and the Lamb shift; for 21 cm the Breit, recoil and proton-structure terms; for ESR the bound g-factor correction $-\alpha^2/3$. Measured values: NIST ASD (vacuum wavelengths), Hellwig et al. (1970), Tiedemann & Robinson (1977), and the established ≈ 10 969 MHz fine-structure interval (e.g. Hagley & Pipkin 1994).

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Fig. S2 — Hydrogen's lines found by kronspice and the model, against measurement

difference from the measured value in parts per million, placed by frequency; the bands show where each line is observed

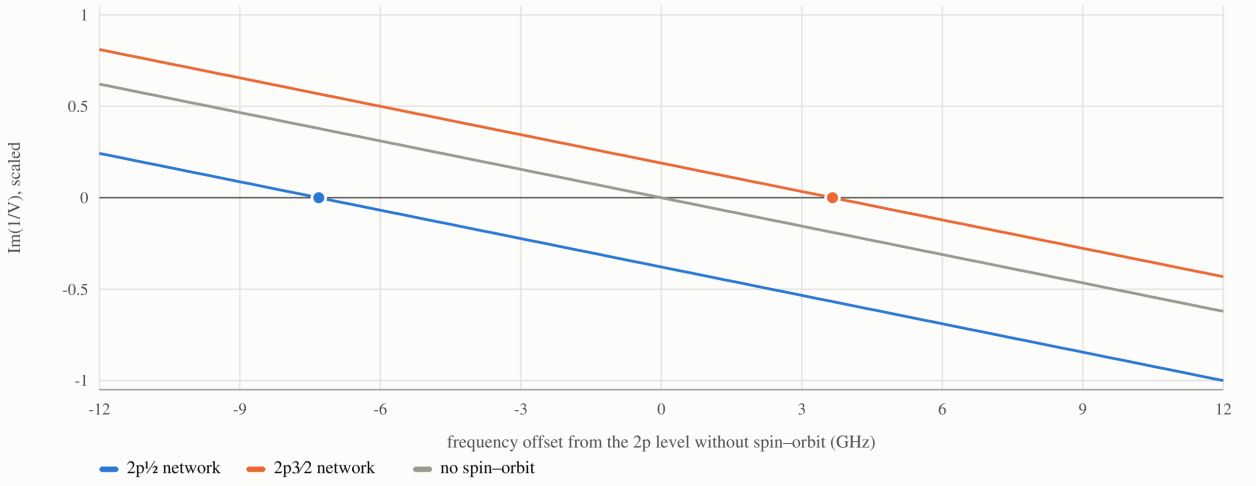


All within 60 ppm. The remainders are physics the models leave out: the Dirac level shifts and the Lamb shift for the optical lines; Breit, recoil and proton-structure terms for 21 cm; the bound-state g-factor correction ($-\alpha^2/3$) for ESR. The ESR point is the frequency per tesla, plotted at 1 T.

Figure 10 (S2). Each line's difference from its measured value in parts per million, placed by frequency from the radio to the ultraviolet. Blue: kronspice circuit resonances; orange: this example's model (Part 1). The ESR point is the frequency per tesla, plotted at 1 T. All lie within 60 ppm.

Fig. S1 — The fine structure as two kronspice resonances

Im(1/V) at the drive of Kron's 2p network (h = 0.05 a0) with the spin–orbit term for j = ½ and j = 3⁄2; each crosses zero at its level



The two zero crossings are 10971.6 MHz apart at this grid; Richardson over h = 0.1 and 0.05 a0 gives 10968.9 MHz, measured 10 969.1 MHz.

Figure 11 (S1). The fine structure as two kronspice resonances: Im(1/V) at the drive of Kron's 2p network (h = 0.05 a0) against the frequency offset from the 2p level without spin–orbit. The j = ½ and j = 3/2 networks cross zero 10971.6 MHz apart at this grid; without spin–orbit the crossing lies between them.

Molecule	kronspice ω_e (cm^{-1})	Measured ω_e	Difference
H_2	3805.0	4401.21	−13.5 %
HD	3295.5	3813.15	−13.6 %
D_2	2691.2	3115.50	−13.6 %

- The Heitler–London spring is too soft: every vibration comes out 13.5 % low, the same error that leaves its bond energy at two thirds of the measured value.
- The isotope ratios, set by the masses alone, agree to 0.07 %: HD/H_2 0.8661 against 0.8664, D_2/H_2 0.7073 against 0.7079; the small remainders are anharmonicity.

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- Rotation B = 1/(2μR²): 44.30 cm^{-1} at the Heitler–London R_e, 60.89 at the measured R_e, against measured B_e 60.853. The bond length, not the formula, is off.

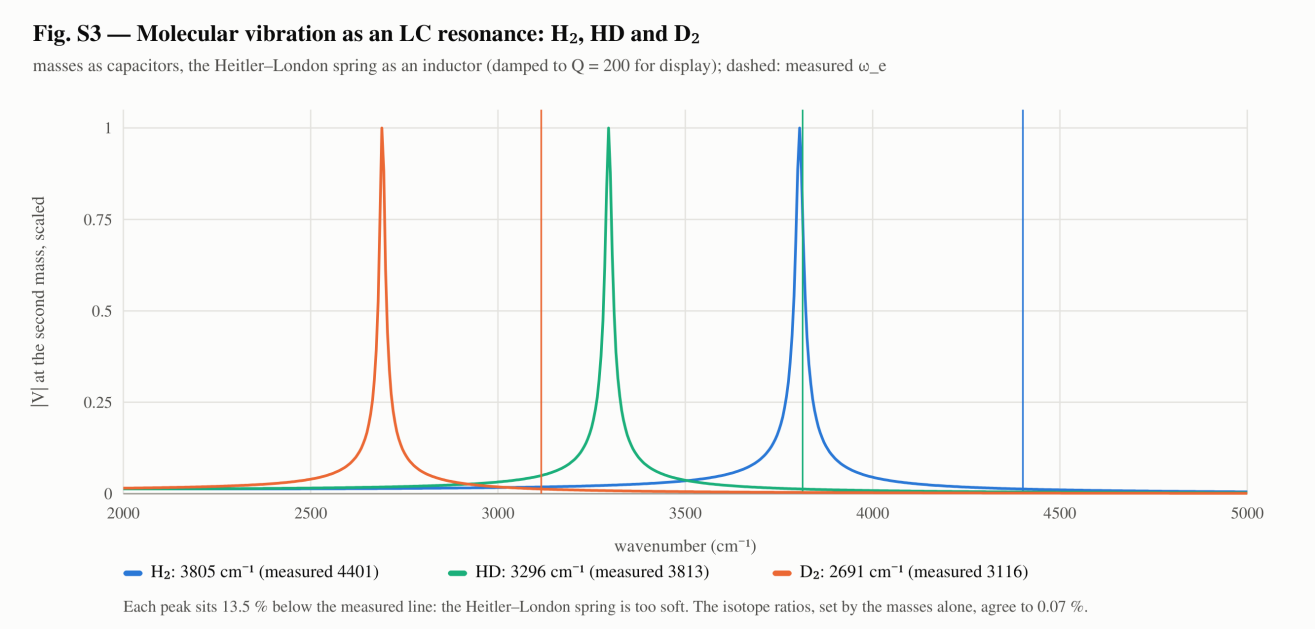


Figure 12 (S3). The vibration circuits of H_2, HD and D_2 (damped to Q = 200 for display): |V| at the second mass against wavenumber. Each peak sits 13.5 % below its measured line (vertical lines).

None of these lines involves the 2p3/2 octupole of Report 1: its magnetic interactions with the proton are far below these intervals, and the proton has no octupole to couple to.

Validation

Test	Checks	Tolerance
TestNormAndDipole	$\int\|\psi\|^2 = 1$; dipole = $-g_j m_j \mu_B$, all four levels	10^{-6}
TestOctupoleSelection	even-l moments zero; j = ½ octupole zero; j = 3/2 ratio −3	10^{-8}; 10^{-6}
TestSeriesTwin	series field = direct loop sum	10^{-3}
TestZones	1s 2 zones at 1–40 a0; 2p3/2 m = ½ 4 zones to 14 a0, 2 beyond 18 a0; r_c = 15.61 a0	exact; 0.05 a0
TestE1	E1(x) against Abramowitz & Stegun Table 5.1	10^{-12}
TestHeitlerLondon	R_e = 1.6425 a0, D_e = 0.1160 E_h; triplet unbound at every R	0.002 a0, 5×10^{-4} E_h
TestSpinDensity	∫ spin density 0 (singlet), 2 (triplet)	exact; 10^{-3}
TestSpectrum	kronspice resonance = matrix; Lyman α to 30 ppm; fine structure, 21 cm to 100 ppm; ESR to 30 ppm; LC pole = √(k/μ); D_2/H_2 ratio	10^{-12} E_h; as listed; 10^{-9}; 2×10^{-3}
TestMoleculeField	triplet far field → dipole of $-g_s \mu_B$; ortho protons 4 zones at 1 a0; para moment 0	10/r²; exact

Limits of the model

- Non-relativistic Pauli spinors and Bohr-type levels with spin–orbit added; Dirac level shifts, the Lamb shift, Breit, recoil and proton-structure terms and the bound g correction are left out.
- H_2 is the Heitler–London wavefunction: its bond is too long and too weak (1.6425 a0, 3.1557 eV against 1.4011 a0, 4.748 eV) and its spring too soft; the field is evaluated at the measured bond length.
- The rotating molecule's own small moment (ortho, J = 1) is not modelled.

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Status

H34 is *bounded*. One atom: ground-state hydrogen has one pair, and so does every j = ½ level; four pairs (an octupole) first appear in 2p3/2, which with m = ±½ shows four N S N S zones round the atom out to 15.6 a0, from the electron's spin magnetization. Two atoms: the bond forms only with the electron spins opposite, so the electron magnets cancel exactly; the proton magnets, 658 times weaker, remain, with a four-zone pattern for ortho-H_2 and no field for para-H_2. What would move it: an observation of an excited atom's magnetic octupole, or of H_2's nuclear four-zone field; neither has been measured, to our knowledge.

Reproducing the results

```
go run ./examples/hydrogen_poles                # about a minute
go run ./examples/hydrogen_poles -svg viz       # also the figures and animations
go test ./examples/hydrogen_poles
python examples/hydrogen_poles/viz/make_report.py examples/hydrogen_poles/hydrogen_poles_report.pdf
```

Sources: W. Russell, "Analysis of the Octave Bar Magnet" (painting; doc/russell-octave-report.md); J. D. Jackson, *Classical Electrodynamics*, 3rd ed., §5.5–5.6; J. J. Sakurai, *Modern Quantum Mechanics*, §3.8, §3.11; Heitler & London, *Z. Phys.* 44, 455 (1927); Sugiura, *Z. Phys.* 45, 484 (1927); Slater (1963), App. 6; Kołos & Wolniewicz, *J. Chem. Phys.* 49, 404 (1968); G. Kron, *Phys. Rev.* 67, 39 (1945); Bethe & Salpeter (1957); NIST ASD; Hellwig et al. (1970); Tiedemann & Robinson (1977); Hagley & Pipkin (1994); Huber & Herzberg (1979); Abramowitz & Stegun, Table 5.1; CODATA 2018.

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